

# RIYA PATEL

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

## EDUCATION

### B.Tech: Computer Science & Engineering

2023 – 2027

P P Savani University, Surat, Gujarat

Coursework: Data Structures & Algorithms · DBMS · Operating Systems · Computer Networks · OOP

## SKILLS

- **Languages:** Python, Java, JavaScript (ES6+), HTML, CSS, SQL
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Backend:** FastAPI, REST APIs, SQLAlchemy, Distributed Systems
- **Databases:** PostgreSQL, Redis, SQL
- **Tools:** Docker, Git, AWS (EC2, S3, RDS, IAM)

## PROJECTS

### WorkerGrid — Distributed Task Processing System — [github.com/Riyadpatel24/workergrid](https://github.com/Riyadpatel24/workergrid)

- *Python · FastAPI · Redis · PostgreSQL · Docker · SQLAlchemy*
- Built a distributed task processing system using FastAPI, Redis, PostgreSQL, and Docker for asynchronous background job execution.
- Designed a producer-consumer architecture with multiple worker processes for scalable task distribution.
- Implemented automatic retries, Dead Letter Queue (DLQ), worker heartbeats, and monitoring APIs for fault tolerance.
- Developed system metrics and task monitoring endpoints to track worker health and task execution.

### Adaptive System — Reliability Monitoring & Adaptive Recovery Platform — [github.com/Riyadpatel24/adaptive\\_system](https://github.com/Riyadpatel24/adaptive_system)

Reliability monitoring platform with real-time telemetry, failure prediction, and fault-injection testing.

- Built a recovery loop: telemetry ingestion → signal normalisation → health classification → LOCKDOWN/THROTTLE actions via safety guard and cooldown manager
- Failure prediction from rolling risk histories and root cause analysis via DependencyGraph; PolicyEngine adapts timeout and retry limits based on live failure rates
- Implemented fault-injection testing using CPU spike and memory leak simulations to validate recovery policies and health classification under failure conditions; FastAPI daemon thread serves live snapshots at `/snapshot` for real-time observability

### VoiceNotes Organizer — AI Audio-to-Notes Pipeline — [github.com/Riyadpatel24/voicenotes-organizer](https://github.com/Riyadpatel24/voicenotes-organizer)

Full-stack pipeline: audio upload → Whisper transcription → LLM-structured notes, tasks, and decisions.

- React + FastAPI pipeline: audio transcribed via Groq Whisper Large V3, then structured by LLaMA 3.3 70B into summaries, action items, deadlines, and decisions in JSON
- Modular architecture cleanly separates transcription and structuring layers — AI models swappable without changing any downstream logic

## CERTIFICATIONS

### IBM Java Developer Professional Certificate

Jun 2025

Coursera / IBM · 11 courses · Spring Boot, Microservices, Docker, REST APIs, NoSQL/SQL, Generative AI

### AWS Academy Graduate — Cloud Foundations

Nov 2025

Amazon Web Services · 20 hours · EC2, S3, RDS, IAM, VPC, cloud security fundamentals

## LANGUAGES

**English:** Professional (C1) | **Hindi:** Native | **Gujarati:** Native | **German:** Elementary (A1) | **Mandarin Chinese:** Elementary (A2)